PRO MINDS

Algebra 1 — Point-Slope Form

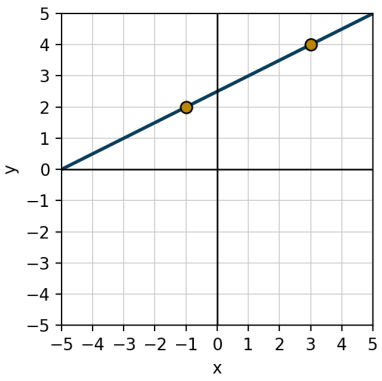
Writing equations in point-slope form from a graph, a table, two points, and a word problem — and converting to slope-intercept form · 10 questions

Name: _____ Date: _____

Remember: Point-slope form is $y - y_1 = m(x - x_1)$, where m is the slope and (x_1, y_1) is a point on the line. Slope: $m = (y_2 - y_1) / (x_2 - x_1)$. Watch the signs — $(x - (-3))$ becomes $(x + 3)$.

Free practice worksheet — circle one answer for each question, show your work underneath, then check yourself with the answer key.

1. Write an equation in point-slope form that models the relationship shown in the graph.



A. $y - 4 = 2(x - 3)$

B. $y - 4 = (1/2)(x - 3)$

C. $y - 3 = (1/2)(x - 4)$

D. $y - 4 = -(1/2)(x - 3)$

2. A table modeling a linear relationship is shown.

x	1	2	3	4
y	7	4	1	−2

What is an equation in point-slope form that models the relationship shown?

A. $y - 4 = -(1/3)(x - 2)$
B. $y - 2 = -3(x - 4)$
C. $y - 4 = 3(x - 2)$
D. $y - 4 = -3(x - 2)$

3. Write an equation of a line that passes through (2, 5) and (5, 11) in point-slope form.

A. $y - 5 = 2(x - 2)$
B. $y - 5 = (1/2)(x - 2)$
C. $y - 2 = 2(x - 5)$
D. $y - 5 = -2(x - 2)$

4. Convert the equation $y + 3 = -4(x - 2)$ into slope-intercept form.

A. $y = -4x + 11$
B. $y = -4x - 5$
C. $y = -4x + 5$
D. $y = 4x + 5$

5. Convert the equation $y - 6 = 3(x + 1)$ into slope-intercept form.

A. $y = 3x + 3$
B. $y = 3x + 9$
C. $y = 3x - 9$
D. $y = -3x + 9$

6. A landscaping company charges a one-time fee plus an hourly charge. For 2 hours of work, the cost is $85, and for 5 hours, it is $175. Write equations that represent the total cost in point-slope and slope-intercept forms.

A. $y - 85 = -30(x - 2)$ and $y = -30x + 25$

B. $y - 175 = 30(x - 5)$ and $y = 30x - 25$

C. $y + 85 = 30(x + 2)$ and $y = 30x - 25$

D. $y - 85 = 30(x - 2)$ and $y = 30x + 25$

7. Write an equation of a line that passes through $(-2, 3)$ and $(2, -1)$ in point-slope form.

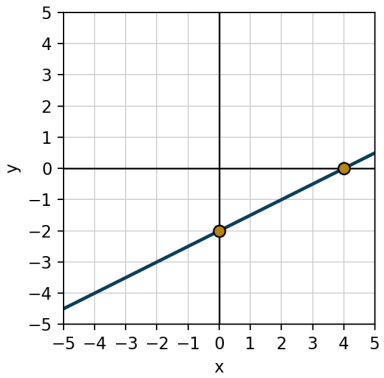
A. $y - 3 = (x + 2)$

B. $y - 2 = -(x + 3)$

C. $y - 3 = -(x + 2)$

D. $y - 3 = -(x - 2)$

8. Write an equation in point-slope form that models the relationship shown in the graph.



A. $y - 0 = (1/2)(x - 4)$

B. $y - 0 = 2(x - 4)$

C. $y + 4 = (1/2)(x - 0)$

D. $y - 4 = (1/2)(x - 0)$

9. A table modeling a linear relationship is shown.

x	0	1	2	3
y	2.5	4	5.5	7

What is an equation in point-slope form that models the relationship shown?

A. $y - 4 = (2/3)(x - 1)$

B. $y - 1 = (3/2)(x - 4)$

C. $y - 4 = -(3/2)(x - 1)$

D. $y - 4 = (3/2)(x - 1)$

10. Write an equation of a line with slope -2 that passes through $(-3, 4)$ in point-slope form.

A. $y - 4 = -2(x - 3)$

B. $y - 4 = -2(x + 3)$

C. $y + 4 = -2(x + 3)$

D. $y - 4 = 2(x + 3)$